ARM PrimeCell

Vectored Interrupt Controller (PL192)

Release Note

PL192_RELN_0000_C

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ARM PrimeCell Vectored Interrupt Controller (PL192) Release Note

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Feedback on the ARM PrimeCell Vectored Interrupt Controller

If you have any comments or suggestions about this product, please contact your supplier giving:

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Change history

Issue	Date	Change
A	28 October, 2002	First release.
B	5 December 2002	Limited Release
C	17th March 2003	Final Release

Contents

1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

Table 1.1-1 lists the ARM part numbers for the individual deliverables that form part of the delivery for this ARM PrimeCell peripheral:

Product deliverables for the ARM PrimeCell PL192 peripheral	Document number	Product code
Synthesizable VHDL RTL	N/A	PL192-MN-23001-r0p0-00rel0
Synthesizable Verilog RTL	N/A	PL192-MN-22001-r0p0-00rel0
Synthesis Scripts (Synopsys Design Compiler, Avant! CB18 cell library)	N/A	PL192-MN-01002-r0p0-00rel0
Testbench - Functional Verification, VHDL Testbench - Integration test, VHDL	N/A N/A	PL192-MN-23010-r0p0-00rel0 PL192-MN-23012-r0p0-00rel0
Testbench - Functional Verification, Verilog	N/A	PL192-MN-22010-r0p0-00rel0
Testbench - Integration test, Verilog	N/A	PL192-MN-22012-r0p0-00rel0
Test vectors-BusTalk, functional verification	N/A	PL192-VE-01010-r0p0-00rel0
Test vectors-BusTalk, integration test	N/A	PL192-VE-01012-r0p0-00rel0
ARM PrimeCell Vectored Interrupt Controller (PL192) Technical Reference Manual (PDF)	ARM DDI 0273 A	PL192-DA-03001-r0p0-00rel0
ARM PrimeCell Vectored Interrupt Controller (PL192) Design Manual (PDF)	PL192 DDES 0000 A	PL192-DC-02002-r0p0-00rel0
ARM PrimeCell Vectored Interrupt Controller (PL192) Integration Manual (PDF)	PL192 INTM 0000 A	PL192-DC-10002-r0p0-00rel0
ARM PrimeCell Vectored Interrupt Controller (PL192) Release Note (PDF)	PL192 RELN 0000 C	PL192-DC-06002-r0p0-00rel0
ARM PrimeCell Vectored Interrupt Controller (PL192) Errata Notice (PDF)	PL192 DEI 0000 A	PL192-DC-11001-r0p0-00rel0
ARM PrimeCell Vectored Interrupt Controller (PL192) Specific Userguide	PL192 USGD 0000 A	PL192-DC-08001-r0p0-00rel0
Software Code Module	N/A	PL192-SW-02001-r0p0-00rel0
Software Test Report	PL192_TREP_1000	PL192-DC-01001-r0p0-00rel0
Software API Document	PL192_ISPC_1000_D	PL192-DC-02001-r0p0-00rel0

The part numbers starting with PL00 below are delivered in separate bundle, part number PL000-BU-00004.

System Common and System Indirection Source Code	N/A	PL000-SW-02001-r9p0-00rel0
System Test Source Code	N/A	PL000-SW-02003-r9p0-00rel0

System Common Files Source Code	N/A	PL000-SW-02004-r9p0-00rel0
System Build Project Files	N/A	PL000-SW-02005-r9p0-00rel0
Interrupt Controller Object Code	N/A	PL001-SW-00000-r9p0-00rel0
Timer Object Code	N/A	PL002-SW-02001-r9p0-00rel0
System Structure and Integration Guide	PL000_DII_1000	PL000-DC-02000-r9p0-00rel0
System Common and System Indirection API Documents	PL000_ISPC_1001 PL000_ISPC_1002	PL000-DC-02001-r9p0-00rel0
System Test API Documents	PL000_ISPC_1003	PL000-DC-02003-r9p0-00rel0
System Common Files API Documents	PL000_ISPC_1004	PL000-DC-02004-r9p0-00rel0
Interrupt Controller API Documents	PL001_ISPC_1000	PL001-DC-02001-r9p0-00rel0
Timer API Documents	PL002_ISPC_1000	PL002-DC-02001-r9p0-00rel0

Table 1.1-1: ARM part numbers for ARM PrimeCell VIC PL192

Note Only the required deliverables will be supplied, as per the contractual agreement.

1.2 Reference Cell Library

A reference cell library (Avant! CB18 0.18µm standard cells) is supplied with this peripheral.

PL000-BU-00005-0.00/Avanti/CB18_2000.3.tar.gz

PL000-BU-00005-0.00/Avanti/CB18_2000.3.lst

1.3 Common driver bundle

A bundle containing software drivers, common to all PrimeCells is also supplied.

This bundle is part PL000-BU-00004 and is supplied as a single tarred and zipped file. The version required is

PL000-BU-00004-r9p0-00bet0 or above.

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX zipped tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of zipped tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions cover the following platform/operating system:

- UNIX

2.2 Disk space required

This installation will require 40 Mbytes of free disk space.

2.3 Installation procedure

You will have a single tarred, zipped and encrypted file of following format:

E.g.

Company Name-PrimeCell-PL192–Bundled-arm-download-WWWW.tgz.des

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the separate products delivered in this shipment.

The first three steps may have already been done, ie the shipment from ARM may have already been decrypted and unzipped. If this is the case then go to step 4, else start with step 1.

1. **Relocate the shipment file**

   Relocate the encrypted file (for example <file>.tgz.des, if DES encryption has been used) to an appropriate location that has sufficient free disk space if PGP encrypted will need the relevant keys.

2. **Decrypt file**

   Follow the accompanying instructions for decryption of the file supplied, to generate a decrypted, compressed, tar file <file>.tgz

3. **Uncompress file**

   Execute the UNIX unzip command to obtain the tar file <file>.tar:

   gunzip <file>.tgz

   where <file> is the name of the file supplied.

4. Extract tar files

To extract the PrimeCell, execute the UNIX tar extract command:

```
tar -xvf <file>.tar
```

This will generate the following three files

ARM_MANIFEST_XXXX.TXT – which lists the PrimeCell files delivered

ARM_DELIVERY_XXXX.TXT – which lists the ARM parts delivered

PL192-DC-06002 -- Release Notes

And will unpack the PrimeCell into the top-level bundle directory PL192-BU-00000-r0p0-00rel0

Beneath this bundle directory are 2 sub directories:

PL192-BU-00000-r0p0-00rel0/PL192_VC contains the PrimeCell development environment.

PL192-BU-00000-r0p0-00rel0/PL192_SW contains software for the PrimeCell

The peripheral development environment PL192-BU-00000-r0p0-00rel0/PL192_VC is divided into two directory structures, a peripheral specific directory structure and a project global directory structure. The top level directory structures are shown in ***Figure 2-1.***

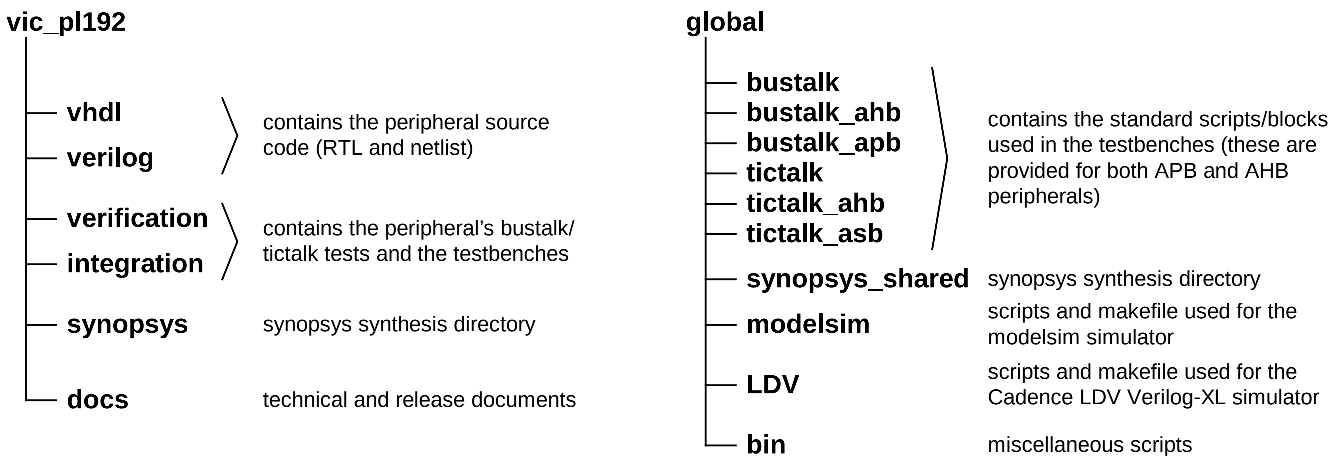

Figure 2-1. Peripheral Development Environment Directory Structure

2.3.2 Installation of the reference cell library

The reference cell library should be copied manually and tar extracted in the directory containing the PrimeCell peripherals. An example is shown below to copy the Avant! CB18 2000.3 cell library to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PL192_PrimeCell/PL000_00005-0.00/Avanti/CB18_2000.3.tar.gz .
```

Execute the UNIX uncompress and tar extract commands as below:

```
gunzip CB18_2000.3.tar.gz
tar -xvf CB18_2000.3.tar
```

2.3.3 Installation of the Common Driver Bundle

The files in the common driver bundle (PL000-BU-00004), unzip and untar into a single directory PrimeCell_gen_sw.

The top level directory structures are shown in ***Figure 2-2.***

```
PrimeCell_gen_sw/
  ├── PL000_DII_1000.pdf
  ├── Documentation/
  └── Software/
        ├── apcommon/
        ├── apint/
        ├── apos/
        ├── aptest/
        ├── aptimer/
        └── build/
```

Figure 2-2 Common PrimeCell Driver Directory structure

3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at http://www.adobe.com. Select “Acrobat Reader" and download the reader for your computer platform/operating system.

3.2 Peripheral Documents Supplied

The following documents are supplied for the ARM PrimeCell Vectored Interrupt Controller PL192:
The document files are located in the vic_pl192/docs/ directory.

ARM PrimeCell Vectored Interrupt Controller PL192
Technical Reference Manual ARM DDI 0273

The technical reference manual provides details for programming and interfacing to this ARM PrimeCell peripheral block.

ARM PrimeCell Vectored Interrupt Controller PL192
Integration Manual PL192 INTM 0000

The integration manual provides information required to integrate this ARM PrimeCell block within a typical industry standard ASIC design flow.

ARM PrimeCell Vectored Interrupt Controller PL192
Design Manual PL192 DDES 0000

The design manual describes the implementation details of this ARM PrimeCell block and the testbenches and test programs used for functional verification.

ARM PrimeCell Vectored Interrupt Controller PL192
Errata Notice PL192 DEI 0000

The errata notice describes details of any known issues with this ARM PrimeCell block and any workarounds for these issues. In most cases this document will be blank or may not be present.

ARM PrimeCell Vectored Interrupt Controller PL192
User Guide PL192 USGD 0000

The user guide describes the instructions on performing simulations and synthesis for this ARM PrimeCell block.

4 TRIAL TESTING

4.1 Product Maturity

This product is newly developed IP and approved for final release.
At the time of release, it has not been implemented in production silicon.

4.2 Technical data summary

The following is a summary of the key technical points from testing of the ARM PrimeCell Vectored Interrupt Controller PL192.

	VHDL RTL	**Verilog RTL**
Cell library	Avant! Passport 2000.3 CB18 v1.0 (0.18 μm)	Avant! Passport 2000.3 CB18 v1.0 (0.18 μm)
Total cell area, without scan, NAND-2 equivalents	33305	31286
Net interconnect area, without scan, NAND-2 equivalents	13447	13469
Total cell area, scan inserted,NAND-2 equivalents	38976	36816
Net interconnect area, scan inserted,NAND-2 equivalents	15890	16248
Clock frequency used	173.91MHz	173.91MHz
Scan insertion figure % (number of 'D' flip-flops replaced)	100% (2007)	100% (2007)
Estimated scan fault coverage, collapsed.	>99%	>99%
Synthesis and scan test tools and version	Synopsys Design Compiler and Test Compiler 2001-08	Synopsys Design Compiler and Test Compiler 2001-08
Simulation tools and versions	Model Technology Inc, ModelSim v5.4a	Model Technology Inc, ModelSim v5.4a Cadence VerilogXL 3.3
Global Peripheral Development Environment Version	Release r8p0-00rel0	

4.3 BusTalk Functional block verification test scenarios

For further details of functional verification testing, refer to the ARM PrimeCell Vectored Interrupt Controller (PL192) Design Manual.

5 DIFFERENCES FROM PREVIOUS REVISIONS

6 KNOWN ISSUES

1. If an access to the VIC is completed in user mode whilst the protection bit in the VICProtection is set (protection mode enabled) the **HRESP[1:0]** will respond with OKAY and not ERROR. The VIC operation is correct, as in a normal system design the user code should not have access to the VIC. The protection mode essentially masks the VIC from user access. Returning an ERROR would require that an extra trap handler is added to the user code.

   In a user system design if DABORT behaviour is required the usual method is to set up the MPU or MMU of the microprocessor core and use its permission check failure to create DABORT.

7 REVISION HISTORY

Date	Revision	Description
28-Oct-2002	r0p0-00bet0	Initial Beta Release 1.0
05-Dec-2002	r0p0-00ltd0	Limited Release 1.0
17th March 2003	r0p0-00rel0	Final Release 1.0